Project Assignment

Analysis of Customer Data

October, 2025

1 Summary

The project assignment will be devoted to predicting player churn using data from a mobile game for Android Dodge the Mud. The goals consist of preparing the dataset, including engineering the necessary features, applying different Machine Learning (ML) algorithms as well as comparing and reporting performance of the models. The theoretical concepts regarding churn prediction will be covered in the first practical session. The deliverables of this assignment will consist of uploading the source code and a written report detailing the workflow and the obtained results on Canvas. While the assignment contributes 20% toward the final grade of this course, it is possible to pass the course without doing the assignment. The assignment however serves as a hands-on learning experience.

2 Dodge the Mud Dataset

Dodge the Mud is a mobile, free-to-play game where the player dodges mud coming from the sky. The dataset was collected between January, 2015 and May, 2016, and is downloadable in a csv format. It contains 153,929 rows and 3 columns: device, score, and time. Variable *device* represents player id, *score* represents the number of muds a player dodged in the game, and *time* refers to timestamp. Timestamps represent seconds and are in Unix Epoch Time format. In its raw form the dataset comes without the outcome variable, making it unsuitable for directly applying supervised ML algorithms. Students are expected to create the outcome variable *churn* (e.g., churner vs non-churner) in addition to engineering other features based on the existing variables *score* and *time*. At least 5 new features, appropriate for the task should be created.

3 Churn

Unlike other business models, where customer churn can be defined by a user terminating their subscription, casual mobile games cannot rely on this metric. The reason for this is that mobile game players seldom delete their accounts. Thus, to obtain the outcome variable to classify a player as a churner or not, some length of player's inactivity should be considered. This is typically done by dividing time into consecutive, non-overlapping periods. These periods are Observation Period (OP) and Churn Prediction (CP) Period, defined in days. It will be up to students to decide how long the OP and CP Periods should be. To create churner vs non-churner label for each player, we consider the following: If a player has one or more play log entries in OP and none in CP Period, then that player is tagged as a churner. Otherwise, if the player shows some activity during CP Period, the player is tagged as a non-churner. Overall, the prepared dataset should contain the newly engineered features, and as a last column in the dataset, the outcome variable churn vs non-churn should be created. Students are also required to check for class imbalance in the dataset and address it if necessary.

4 Churn Prediction Modeling

The modeling part of this assignment consists of modeling churn prediction using three ML algorithms, i.e., Logistic Regression (LR), Extreme Gradient Boosting (XGBoost), and Random Forest (RF). For each model, students should perform cross-validation to evaluate the performance. The performance results should be compared using performance measures that are deemed appropriate for the problem at hand.

5 Assignment Deliverables

5.1 Notebook

Students are requested to submit a Jupyter notebook with the project's solution, including the code and the comments regarding each step.

5.2 Written Report

In addition to the notebook, students are required to submit a written report detailing the process and the results, along with their discussion (maximum 4 pages). The report should cover: (1) the definition of churn, (2) an explanation of the OP and CP Period, including the specific periods chosen for the project, (3) the steps taken to explore and pre-process the dataset, including the process of creating the outcome variable and engineering at least five new features, (4) any feature selection techniques considered or applied, (5) the description of training the models (LR, XGBoost, and RF) and tuning hyper-parameters, (6) a discussion of the model evaluation process using stratified 5-fold cross-validation, (7) a presentation of performance results based on appropriate measures (e.g., ROC-AUC, etc.), (8) a critical discussion and reflection of the obtained results. The written report must use clear academic English and, where needed, adhere to the APA reference format, as well as state what each team member contributed toward completing the project (i.e., in terms of Conceptualization, Methodology, Analysis, Writing - Original draft, Writing - Review & Editing, Visualization). It is also required to include a figure depicting the overall ML workflow of the project (i.e., the different steps taken and the order of their sequence) in the Methods section of the report.

6 Submission and Deadline

The Jupyter notebook and the written report must be uploaded to Canvas. Submissions via email will not be accepted unless some exceptional situation comes to play. There will be **one** submission opportunity on Canvas as indicated by the deadline, **December 5th**.

7 Grade

The project assignment will count for 20% of the course grade, determined numerically (on a scale from 1 to 10). Submitting both a clean and well-commented code solution, along with the written report, is mandatory to pass the assignment. If the code does not run or is incorrect, the grade is 0. Similarly, if the written report is missing, the grade is 0. The grade will be evaluated based on the following criteria: (1) Data Preparation (the appropriateness of pre-processing steps, the features created, and the outcome variable created), (2) Model Implementation (correct implementation of the three models and a reasonable discussion on how parameter choices were made), (3) Model Evaluation and Results (proper use of evaluation metrics and the interpretation of model performance), (4) Discussion of Results (the reflection of the trade-offs made in hyper-parameter tuning, etc.), (5) Overall Code and Report Quality (well-structured and commented code, explanations of the steps and the decisions taken in the workflow, adherence to the submission guidelines). You shall also declare your Generative AI usage if any.

8 Working in Groups

Students will work in groups of two (minimum) or three to four students (maximum). Doing the project alone is not allowed, as we aim to promote collaboration among students. While each student is responsible for enrolling in a group, we will provide assistance in case you run into the difficulty of finding a group. Groups shall be registered on Canvas by **October 30th** at the latest.

9 Communication and Questions

The communication of this course will be implemented via the discussion section of Canvas for sharing important announcements and answering questions. Thus, we encourage students to post all questions there and refrain from contacting the instructor of the practical sessions (dr. Julija Vaitonytė) via email. However, feel free to contact the instructor via email if there is a private issue that you feel your fellow students should not be aware of. Importantly, students are not allowed to ask the instructor to revise their projects or provide feedback prior to submission.

9.1 Asking Questions

Please make sure that your questions are to the point and that similar questions have not been posted. Students are encouraged to answer each other's questions. The instructor will always validate the answers and further expand them whenever opportune. Overall, students can expect a reply within three working days after the question is posted on Canvas (during working hours).

10 Generative AI Use

While Generative AI tools can be useful for exploring ideas or clarifying concepts, students are strongly encouraged to complete the main analysis and writing independently. The purpose of the assignment is to enable the use of critical thinking and data science skills, which is undermined when the work is outsourced to a Generative AI tool. Moreover, Generative AI tools may produce errors, thus introducing misleading information into your project.